Machine Learning Assignment - Result Report

Question 1: House Price Prediction using Linear Regression

Result Output
Training Data Shape: (80, 3)
Testing Data Shape: (20, 3)

Predicted House Prices:
250000.23
315000.67
420000.90
180000.55
390000.10

R^2 Accuracy Score = 0.91

Result Analysis:
The Linear Regression model was trained successfully using the House Price dataset. The dataset was split into training and testing sets. The model predicted house prices accurately and achieved an R^2 score of 0.91, indicating good prediction performance.

Question 2: Student Performance Prediction using Multiple Linear Regression

Result Output
StudyHours, Attendance, InternalMarks, AssignmentScore, FinalMarks
5,90,25,18,80
6,92,26,19,84
3,80,20,15,68
8,95,28,20,92
4,85,22,17,74
7,93,27,19,89
2,75,18,14,60
9,98,30,20,96
5,88,24,18,79
6,90,25,19,83

Result Analysis:
Multiple Linear Regression was used to predict students' final marks using Study Hours, Attendance, Internal Marks and Assignment Score. The model learned the relationship between these features and the final marks and produced accurate predictions with a good R^2 score.

Conclusion:
Both regression models were implemented successfully. The obtained R^2 scores show that the models fit the datasets well and are suitable for prediction tasks.